

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Applied Mathematics
Examination: 2nd Periodical Examinations
Branch: M.E/ECE/CSE

Course Code: MA-111
B.Tech Scheme: NEP
Semester: 1st

Question Paper for 2nd Periodical Examination (NOV 2025)

Time: One and half Hours

Max Marks: 30

Note: Attempt all questions

CO No.	Course Outcomes
CO.1	Examine linear simultaneous equations for consistency and applications
CO.2	Apply concept of linear algebra to convert quadratic form to canonical forms
CO.3	Solve problems on improper integral & Inspect the function of two variables for optimization.
CO.4	Evaluate double and triple integrals in various coordinates & their application in engineering
CO.5	Test for the analyticity of complex functions

Q.N o.	Questions	Marks	CO	BL
1.a	Use Taylor's theorem to find $\sqrt{66}$	1	CO1	L1
1.b	Find first and second order partial derivatives of $Z = X^3 + Y^3 - 3aXY$	1	CO4	L1
1.c	Define the Euler theorem	1	CO4	L1
1.d	Show that $\partial(u,v)/\partial(x,y) = [\partial(u,v)/\partial(r,s)] \times [\partial(r,s)/\partial(x,y)]$	1,1	CO4	L2
2.	Prove that $\log(1+\sin x) = x - x^2/2 + x^3/6 - x^4/12 + \dots$ by using Taylor's series	5	CO4	L3
3.	If $u = \log(X^3+Y^3+Z^3-3XYZ)$, prove that $= 9(X+Y+Z)^{-2}$	5	CO1	L3
4.	Find the minimum value of $f = x^2 + y^2 + z^2$ given that $ax + by + cz = p$	5	CO1	L5
5.	Evaluate the following integral: $\int (x = 0 \text{ to } 4a) \int (y = x^2/4a \text{ to } 2\sqrt{ax}) dy dx$	10	CO4	L5

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